

Medical DB Query Generation for SR

Currently: Seed papers $\Rightarrow$ Key terms $\longrightarrow$ come up with query
use strategies, combination of keywords & their variations
 $\longleftarrow$ trial & error
w/ multiple queries (combination of a bunch)

Goal: $\longrightarrow$ get concise studies, yet don't overfetch 

Initial Approach:

Step 1) Get keywords $\longrightarrow$ SR into LLM to get PICO
 $\hookrightarrow$ Summarize $\longrightarrow$ TFID or some kind of similarity word search

```
1 "Myocardial Ischemia"[Mesh] OR "Coronary Artery Disease"[Mesh] OR
  "Acute Coronary Syndrome"[Mesh] OR "Myocardial Infarction"[Mesh] OR
  "Angina Pectoris"[Mesh] OR "Percutaneous Coronary Intervention"[Mesh]
  OR "Coronary Artery Bypass"[Mesh]

2 Myocardial Ischemia[Title/Abstract] OR Ischemic Heart
  Disease[Title/Abstract] OR Heart Disease*[Title/Abstract] OR Heart
  attack[Title/Abstract] OR Coronary Heart Disease[Title/Abstract] OR
  Coronary Artery Disease[Title/Abstract] OR Coronary Disease*[Title/Ab-
  stract] OR Acute Coronary Syndrome[Title/Abstract] OR Acute Coronary
  Syndrom*[Title/Abstract] OR Myocardial Infarction[Title/Abstract] OR
  Angina*[Title/Abstract] OR Percutaneous Coronary
  Intervention[Title/Abstract] OR Percutaneous Coronary Revascularization*
  [Title/Abstract] OR Percutaneous Transluminal Coronary Angioplast*
  [Title/Abstract] OR Coronary Balloon Angioplast*[Title/Abstract] OR
  Coronary Artery Bypass*[Title/Abstract]

3 #1 OR #2

4 ("Rehabilitation"[Mesh]) OR (Rehabilitation[Title/Abstract] OR
  rehab[Title/Abstract])

5 "Telemedicine"[Mesh] OR mhealth[Title/Abstract] OR m-health[Title/Ab-
  stract] OR mobile health[Title/Abstract] OR mobile application[Title/Ab-
  stract] OR mobile device[Title/Abstract] OR mobile communication*
  [Title/Abstract] OR mobile phone[Title/Abstract] OR smartphone[Title/Ab-
  stract] OR smart phone[Title/Abstract] OR smartphone application[Title/Ab-
  stract] OR cellphone[Title/Abstract] OR cellular phone[Title/Abstract] OR
  telemedicine[Title/Abstract] OR telerehabilitation[Title/Abstract] OR tele-re-
  habilitation[Title/Abstract] OR Virtual Rehabilitation[Title/Abstract] OR
  Remote Rehabilitation[Title/Abstract] OR telehealth[Title/Abstract] OR tele-
  health[Title/Abstract] OR telemonitor[Title/Abstract] OR ehealth[Title/Ab-
  stract] OR e-health[Title/Abstract] OR digital health[Title/Abstract] OR mo-
  bile technolog*[Title/Abstract] OR website[Title/Abstract] OR web-
  based[Title/Abstract]
```

6 3 and 4 and 5

Step 2) Make queries.
 $\Rightarrow$ Common method:

bunch of OR w/ initial searches
 $\rightarrow$ then aggregate everything with
AND from previous results.

Step 3)

Approach 2) Include studies in bibliography. w/ seed papers.

Step 1) Fetch all papers referenced, even some not in "include.xlsx" & parse them

Step 2) Build PICO for SR itself & compare relevance against all references using a similarity model. Filter all studies below certain threshold.

Step 3) W/ Filtered studies parse their abstracts if available & get relevant terms.

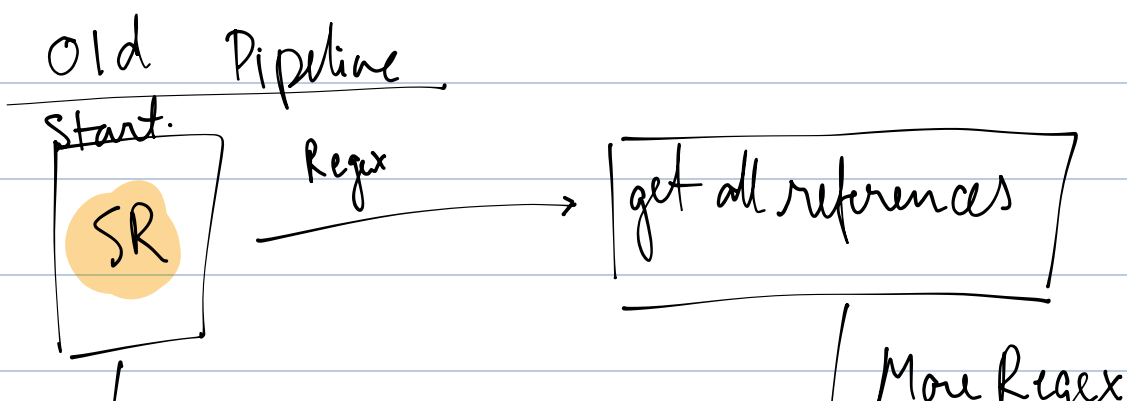
from pico → look for terms

⇒ Insight: 1) To get sensitive list: use precise terms
e.g. renal failure vs kidney

2) Get good accuracy / good recall. → include all term synonyms
Overall → include all synonym terms within PICO and be precise. no vague term (they give the shit results)

Article I found on LLMs making queries:

<https://academic.oup.com/jamiaopen/article/7/3/ooae098/7775509>



LLM

Get title or DOI if available

Get PICO Facets
→ Mainly population & Intervention

Run queries on PubMed AP

Reference Metadata
(Not so good: has a lot of missing return values)

→ cached into JSON file

Gemini

TF-IDF

Cosine similarity search to root out useless references:
Final relevant text

Gemini

Extract Keywords either MESH or free text.

From PICO, Gem. extracts n key words from population & interventions (e.g. 3)

Gemini

try to filter

Extracted terms get refined to make more relevant to "Core idea" with keywords

with final filtered words

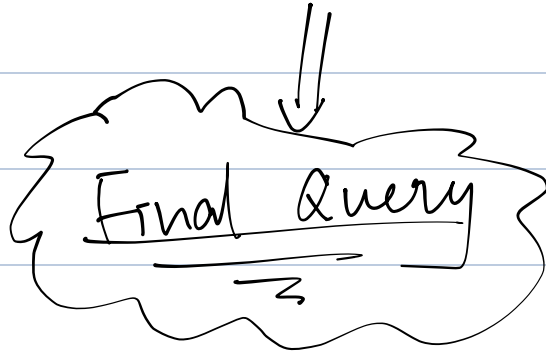
Gemini

→ make query: mix of Mesh and free text

ex: (pop_1 or pop_2 or) and (inter_1 or inter_2 or ...) and RCT

cuts results by a lot

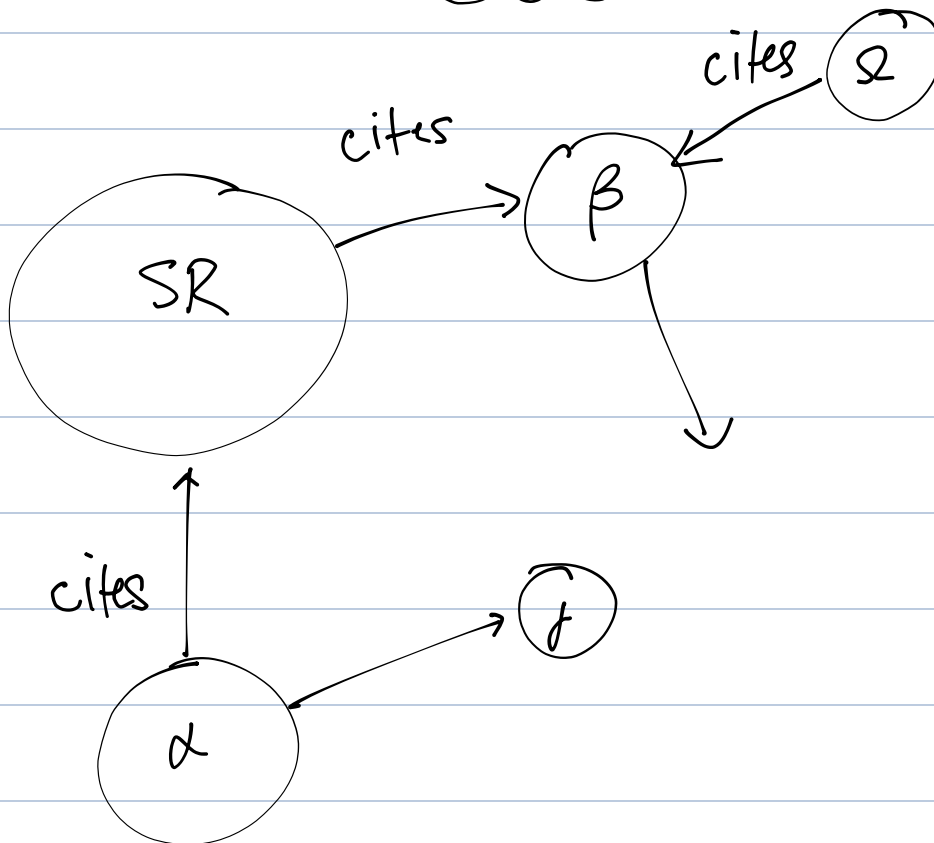
} randomized control trial
(if SR - deems it so)



Observations: used gemini because of free credits.

⇒ results vary, good ones include ~ 900 results for 88% recall

New Approach (Graph Method)



Final result is big as adjacency list to better

represent connections between article = better gauge for relevancy + covers more literature

Issues w/ previous method:

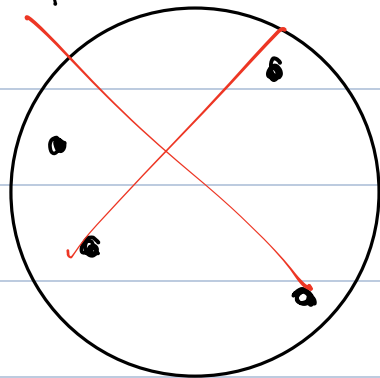
1) - Too precise or too vague. → filtering wrong words

2) → some articles don't have MESH

→ Exceptions for lack of recall, some articles not in PubMed, some DOI don't work, etc...

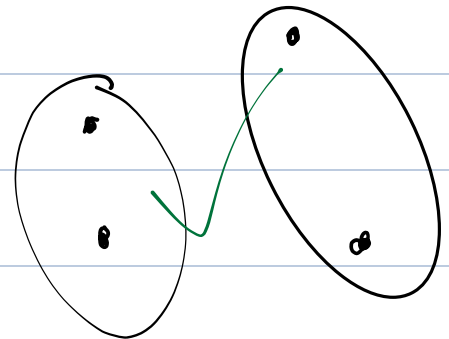
⇒ **Solution:** More context: BFS to make adjacency list w/ citations + adding more variant to core words.

Result: Get more specific terms & their variant so



Relying on Mesh words

instead



less noise with free words.

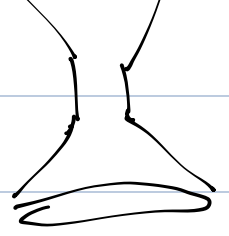
Insights: → Discovered that MeSH terms are very unpredictable, slight change can lead to amount of results to explode. ⇒ Conclusion: Make MESH terms quite specific in query and add more variants and more synonyms in free terms.

⇒ Basic Idea: Get as much context as possible, filter ideas, then broaden terms

New Improved Pipeline like "Double funnel" →

Start
SR

Regex → get all references



LLM

Get PICO Facets
→ Mainly population & Intervention

More Regex
Get title or DOI if available

Build Graph

OpenAlex Api ①
First round forward & backwards search on SR PDF

Results get piped

②
2nd round forward & backward search on all references from 1st round

TF-IDF

TF-IDF

Filter by the amount of times getting from ①

Builds Adjacency List cache stored as JSON

Cosine similarity search to root out useless references:
Final relevant text

Geniuni

Find relevant terms
Mesh & free text w/PICO

Find Free
Extra free text related to SR

++

Free term expansion
e.g. add variants/synonyms

focus on free-text.

Final Query